

The Bernoulli D. Equation is an extension to the separable D. Equation

Select one:

- ☐ True
- ☒ False



The differential equation whose general solution $A\cos(2x) + B\sin(2x)$ is

Select one:

☐ a. $y'' - 4y = 0$

☐ b. $y'' + y = 0$

☐ c. $y'' + 2y = 0$

☒ d. $y'' + 4y = 0$

Every separable D. Equation is an Exact D. Equation

Select one:

- ☒ True
- ☐ False



If $y = e^x$ is a solution of $y'' + y' + cy = 0$,
then $c = 2$

Select one:

- ☐ True
- ☒ False



The Laplace transform $L(t^3 \delta(t - 2)) =$

Select one:

- ☐ a. s^{-2s}
- ☐ b. $2s^{2s}$
- ☒ c. $8s^{-2s}$
- ☐ d. $2s^{-2s}$

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$$L\{(t-1)^2\} = \frac{2}{s^3} - \frac{2}{s^2} + \frac{1}{s}$$

Select one:

- ☒ True
- ☐ False



The general solution of the Euler-Cauchy D.
Equation $x^2 y'' - 2xy' - 4y = 0$ is

Select one:

☐ a. $Ax^{-1} + Bx^{-4}$

☒ b. $Ax^{-1} + Bx^4$

☐ c. $Ax^1 + Bx^4$

☐ d. $Ax^{-1} + Bx^2$

The ODE $\frac{dy}{dx} + x = y$ is linear.

Select one:

- ☒ True
- ☐ False

$$L\{sint\} = \frac{1}{s^2-1}$$

Select one:

☐ True

☒ False



The ODE $(x^2 + 1)dx + (y^2 - 1)dy = 0$ is exact.

Select one:

- ☒ True
- ☐ False



$$L^{-1}\left\{\frac{e^{-2s}}{s^2-4}\right\} =$$

Select one:

☒ a. $\frac{1}{2} \sinh(2(t-2))u(t-2)$

☐ b. $\frac{1}{2} \sinh(2(t-1))u(t-2)$

☐ c. $\sinh(2(t-2))u(t-2)$

☐ d. $\frac{1}{2} \cosh(2(t-2))u(t-2)$

The Maclaurin series for $\sin x$ is $\sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$

Select one:

- ☐ True
- ☒ False

While using the power series $y = \sum_{k=0}^{\infty} a_k x^k$ to solve a differential equation we must have $y' = \sum_{k=2}^{\infty} a_k k x^{k-1}$

Select one:

- ☐ True
- ☒ False

The eigen values λ of the system $X' = AX$ can be obtained from the relation $\det(A - \lambda I) = 0$, I being the identity matrix

Select one:

- ☒ True
- ☐ False

The ODE $(x^3 + 1)dx + (y^7 - 1)dy = 0$ is exact.

Select one:

- ☒ True
- ☐ False



$$L\{2(\sin t)(\cos t)\} = \frac{1}{s^2+4}$$

Select one:

☐ True

☒ False

If $f(t) = L^{-1}\left\{\frac{s}{s^2+1}\right\}$, then $f(0) =$

Select one:

- ☐ a. -1
- ☒ b. 1
- ☐ c. 2
- ☐ d. 0



$$L^{-1}\left\{5 + \frac{2}{s-1}\right\} = 5\delta(t-1) + 2e^t$$

Select one:

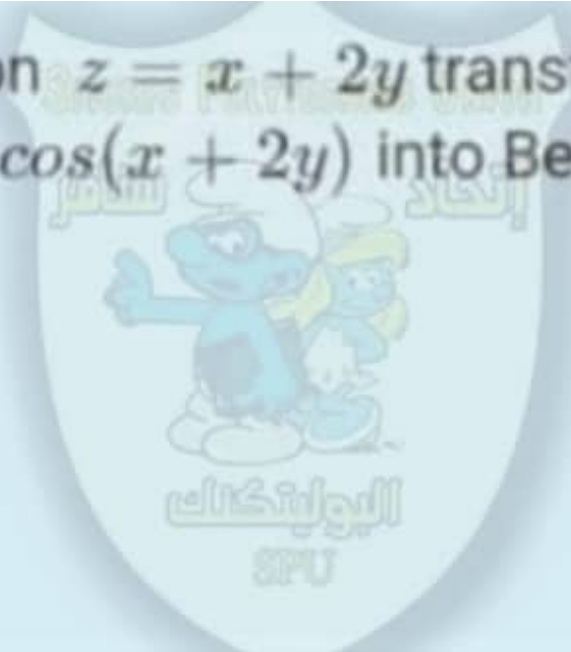
- ☒ True
- ☐ False



The substitution $z = x + 2y$ transforms the equation $y' = \cos(x + 2y)$ into Bernoulli

Select one:

- ☐ True
- ☒ False



If $y''' + 2y'' - 4y' - 8y = e^{-2x}$, then the particular solution $y_p =$

Select one:

☐ a. $Dx^3 e^{-2x}$

☐ b. $Dx e^{-2x}$

☐ c. $(Dx + C)e^{-2x}$

☒ d. $Dx^2 e^{-2x}$

If $\frac{M_y - N_x}{M} = g(y)$, then the integrating factor is

Select one:

- ☐ a. $e^{\int g(y)dy}$
- ☐ b. $e^{\int 4g(y)dy}$
- ☐ c. $2y$
- ☒ d. $e^{-\int g(y)dy}$



If the general solution of
 $y''' - 3y'' + 3y' + ky = 0$ is
 $y = Ae^x + Bxe^x + Cx^2e^x$, then the value of
 k is

Select one:

- ☐ a. -3
- ☐ b. 2
- ☒ c. -1
- ☐ d. 1



If A is a 2×2 matrix with eigenvalues $-4, 4$ and eigenvectors $\vec{K}_1$ and $\vec{K}_2$, then the general solution of the system $X' = AX$ is

$$X(t) = c_1 \vec{K}_1 e^{-4t} + c_2 \vec{K}_2 e^{4t}.$$

Select one:

- ☒ True
- ☐ False

To convert the Bernoulli's equation
 $\frac{dy}{dx} + xy = y^4$ to linear we consider the
substitution $z = y^2$.

Select one:

- ☐ True
- ☒ False

If f is a function and u is a unit step function then the Laplace transform

$$L(f(t - 5)u(t - 5)) =$$

Select one:

☐ a. $e^s L(f)(s)$

☐ b. $e^{-5s} L(f)(s)$

☐ c. $e^{5s} L(f)(s)$

☒ d. $e^{-5s} f(s)$

$$L^{-1}\left\{\frac{s+1}{(s-3)^2+16}\right\} =$$

Select one:

- ☐ a. $e^{3t} \cos(4t) + e^{3t} \sin(4t)$
- ☐ b. $e^{-3t} \cos(4t) + e^{-3t} \sin(4t)$
- ☐ c. $\frac{1}{4}e^{3t} \cos(4t) + e^{3t} \sin(4t)$
- ☐ d. $e^{3t} \cos(4t) + \frac{1}{4}e^{3t} \sin(4t)$

If $L\{f(t)\} = \ln(s + 4) - \ln(s + 6)$, then $\int_0^\infty f(t)e^{-2t}dt =$

Select one:

☐ a. $\ln\left(\frac{5}{7}\right)$

☐ b. $\ln\left(\frac{6}{8}\right)$

☐ c. $\ln\left(\frac{2}{4}\right)$

☐ d. $\ln\left(\frac{7}{5}\right)$

$$L\{\int_0^t \tau \sin(8\tau) d\tau\} =$$

Select one:

☐ a. $\frac{1}{(s^2+64)^2}$

☐ b. $\frac{2}{(s^2+64)^2}$

☐ c. $\frac{2s}{(s^2+64)^2}$

☐ d. $\frac{2s}{(s^2+64)}$

If we use the method of undetermined coefficients to find y_p for $y''' - y'' = 2 + e^x$, then the suitable form for y_p is

Select one:

- ☐ a. $Ax + Bxe^x$
- ☐ b. $Ax^2 + Bxe^x$
- ☐ c. $Ax^2 + Bx^2e^x$
- ☐ d. $Ax^3 + Bxe^x$

One of the following D.E. is linear

Select one:

- ☐ a. $y^2 y' + 3y^2 = x^3$
- ☐ b. $x^2 y'' = 2y^2$
- ☐ c. $xy^2 y' = (y^2 - xy^3)$
- ☐ d. $xy' + (y^2 - x^2 y - xy) = 0$

The D.E. $xy'' - x^2y' + \csc(x)y = \frac{1}{x-3}$ has a solution in one of the following intervals

Select one:

- ☐ a. $(0, 2\pi)$
- ☐ b. $(0, \pi)$
- ☐ c. $(0, 3)$
- ☐ d. $(-\pi, 3)$

The D.E. that has the general solution $y = x^{-3}(c_1 \cos 2 \ln x + c_2 \sin 2 \ln x)$ is

Select one:

☐ a. $x^2 y'' + 7xy' + 13y = 0$

☐ b. $x^2 y'' - 7xy' - 13y = 0$

☐ c. $x^2 y'' + 3xy' + 13y = 0$

☐ d. $x^2 y'' - 3xy' + 13y = 0$

The value of m that makes $y = x^m$ to be a solution to $x^2 y' + 4xy = 0$ is

Select one:

- ☐ a. $\frac{4}{3}$
- ☐ b. -3
- ☐ c. -4
- ☐ d. 2

One of the following is a second order nonlinear D.E.

Select one:

☐ a. $y' + y''x = ye^x$

☐ b. $y' + y \ln y = x$

☐ c. $y + y'' \cos x = \frac{1}{x}$

☐ d. $yy' + y'' \sin x = 1$

Let $y = \sum_{n=0}^{\infty} a_n x^n$ be a series solution to a second order D.E., whose recurrence relation is

$a_{n+2} = \frac{-na_n}{n+2}, n = 0, 1, 2, 3, \dots$, then the general solution is $y =$

Select one:

- ☐ a. $a_0 + \sum_{n=0}^{\infty} \frac{1}{2n+1} a_1 x^{2n+1}$
- ☐ b. $a_0 + \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} a_1 x^{2n+1}$
- ☐ c. $a_0 + \sum_{n=0}^{\infty} \frac{(-1)^n}{n+2} a_1 x^{n+2}$
- ☐ d. $a_0 + \sum_{n=0}^{\infty} \frac{(-1)^n}{2n-1} a_1 x^{2n-1}$

If the variation of parameters is used to find $y_p = y_1 u_1 + y_2 u_2$ for $x^2 y'' - 3xy' + 3y = x^4$ such that $y_c = c_1 x + c_2 x^3$ and the Wronskian of the two solutions is $W = 2x^3$. Then $u_1 =$

Select one:

- ☐ a. $\frac{-1}{10} x^5$
- ☐ b. $\frac{1}{6} x^3$
- ☐ c. $\frac{1}{10} x^5$
- ☐ d. $\frac{-1}{6} x^3$

Suppose that $y(0) = 1$ and $L\{y(t)\} = Y(s)$ such that $L\{e^{2t}y'(t)\} = Y_1(s)$, then $Y_1(4) =$

Select one:

- ☐ a. $6Y(6) - 1$
- ☐ b. $6Y(6) + 1$
- ☐ c. $2Y(2) + 1$
- ☐ d. $2Y(2) - 1$



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The general solution of the exact D.E.

$$(3x^2y + \sec^2 x)dx + (x^3 + 2y)dy = 0 \text{ equals}$$

Select one:

☐ a. $x^3y + y^2 + \tan x = c$

☐ b. $x^3y + y^3 + \tan x = c$

☐ c. $x^3y + y + \tan x = c$

☐ d. $x^2y + y^2 - \tan x = c$

If the roots of the characteristic equation of a H.L.D.E. with constant coefficients are $1, -2$, then the D.E. is

Select one:

☐ a. $y'' + y' + 2y = 0$

☐ b. $y'' - y' + 2y = 0$

☐ c. $y'' - y' - 2y = 0$

☐ d. $y'' + y' - 2y = 0$